

Soliton Spin-Chain Heatmaps: Real Space and Momentum–Energy (k – ω) Space

Heisenberg Spin Chain — Solitons

June 3, 2026

Contents

For each parameter set we show the spin signal in real space (δS over site j and time t) alongside its two-dimensional Fourier transform in momentum–energy space ($|S(k, \omega)|^2$). All runs use a single initial condition (IC = 1).

1 $N = 5$ ($L = 500$)

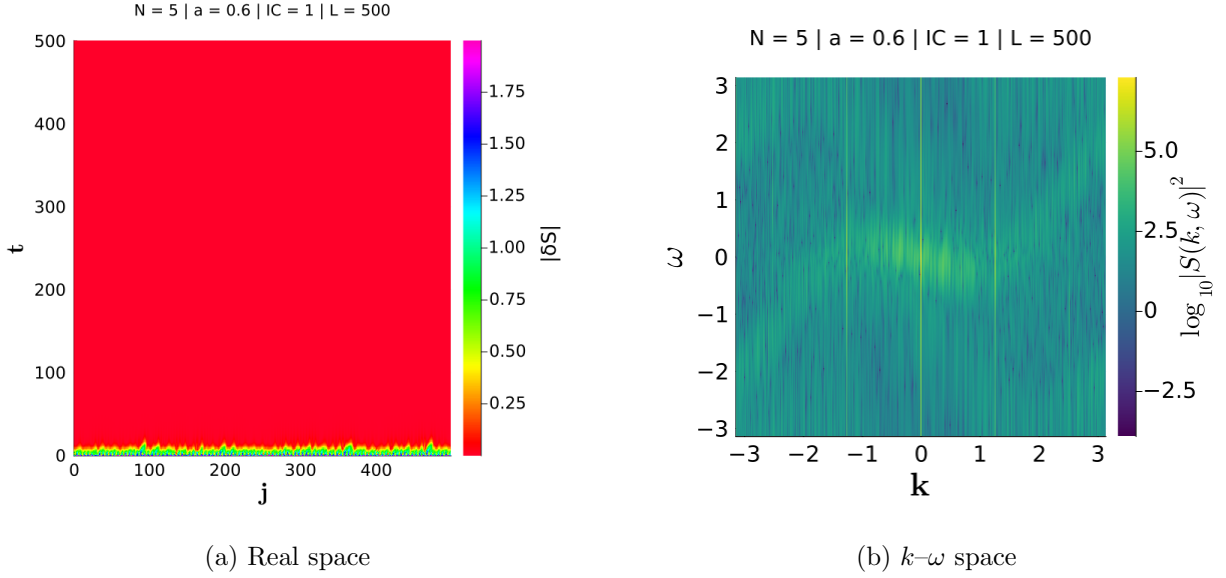


Figure 1: $N = 5$, $a = 0.6$, $L = 500$.

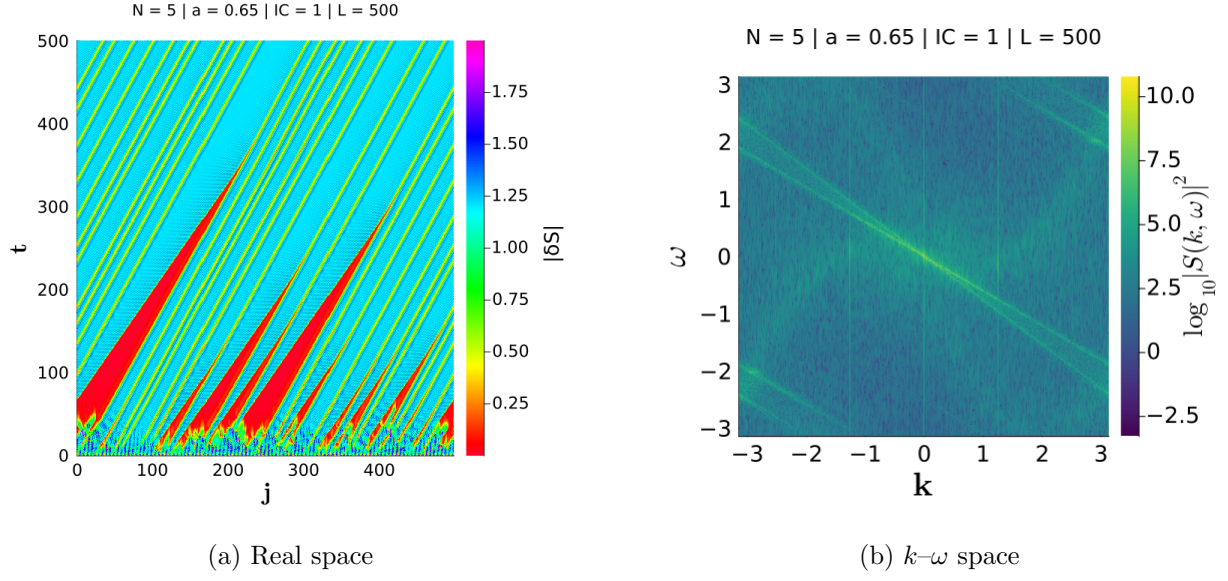


Figure 2: $N = 5$, $a = 0.65$, $L = 500$.

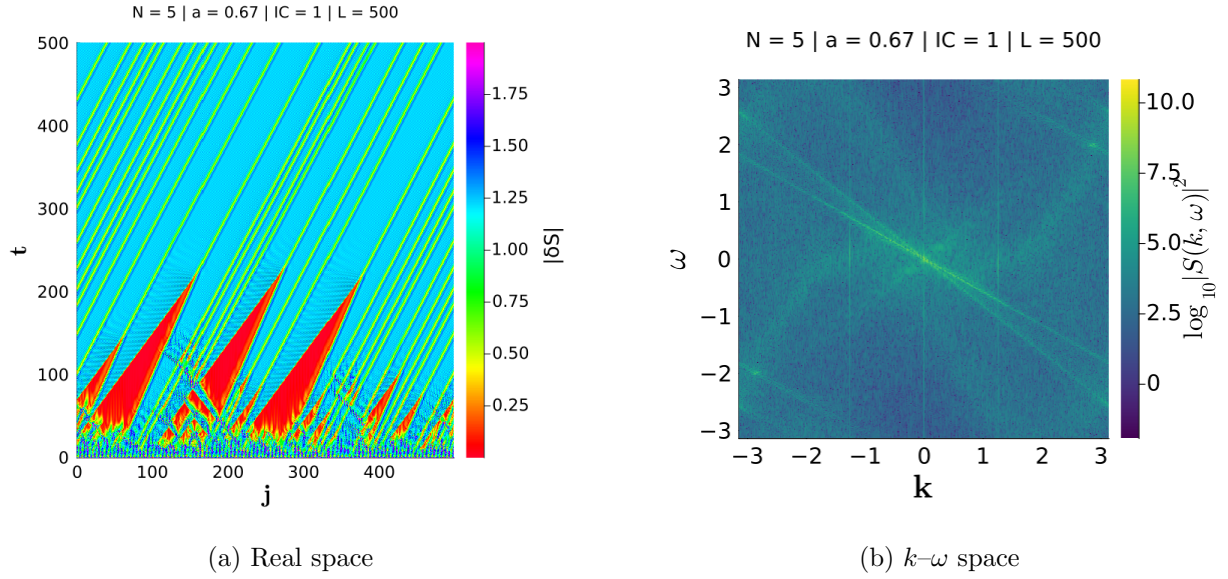
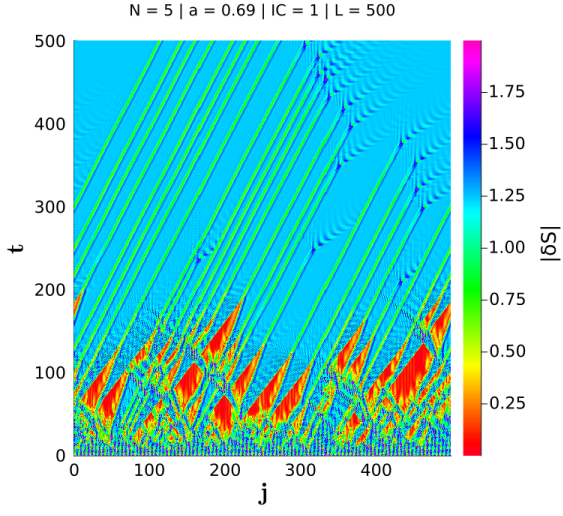
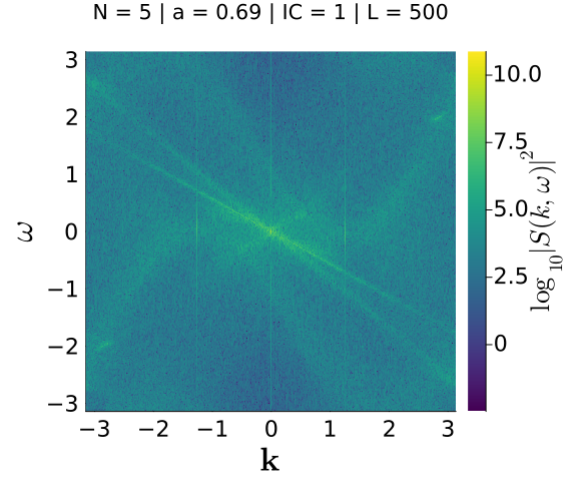


Figure 3: $N = 5$, $a = 0.67$, $L = 500$.

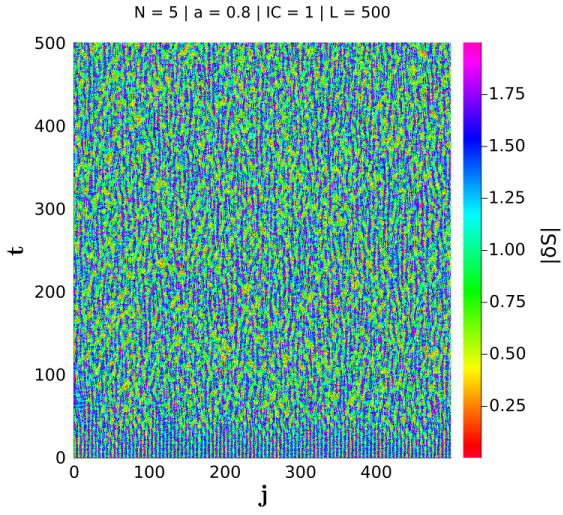


(a) Real space

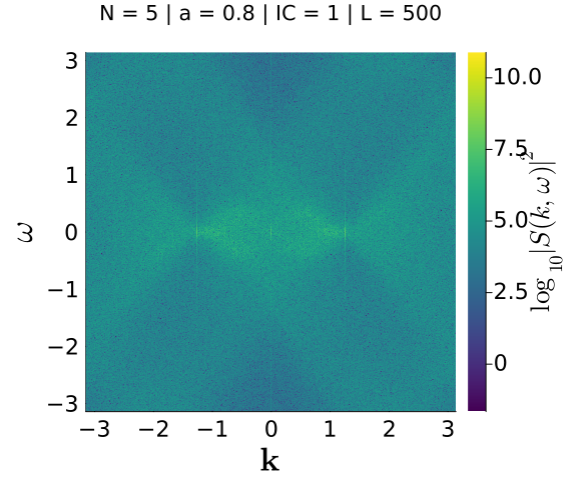


(b) k - ω space

Figure 4: $N = 5$, $a = 0.69$, $L = 500$.



(a) Real space



(b) k - ω space

Figure 5: $N = 5$, $a = 0.8$, $L = 500$.

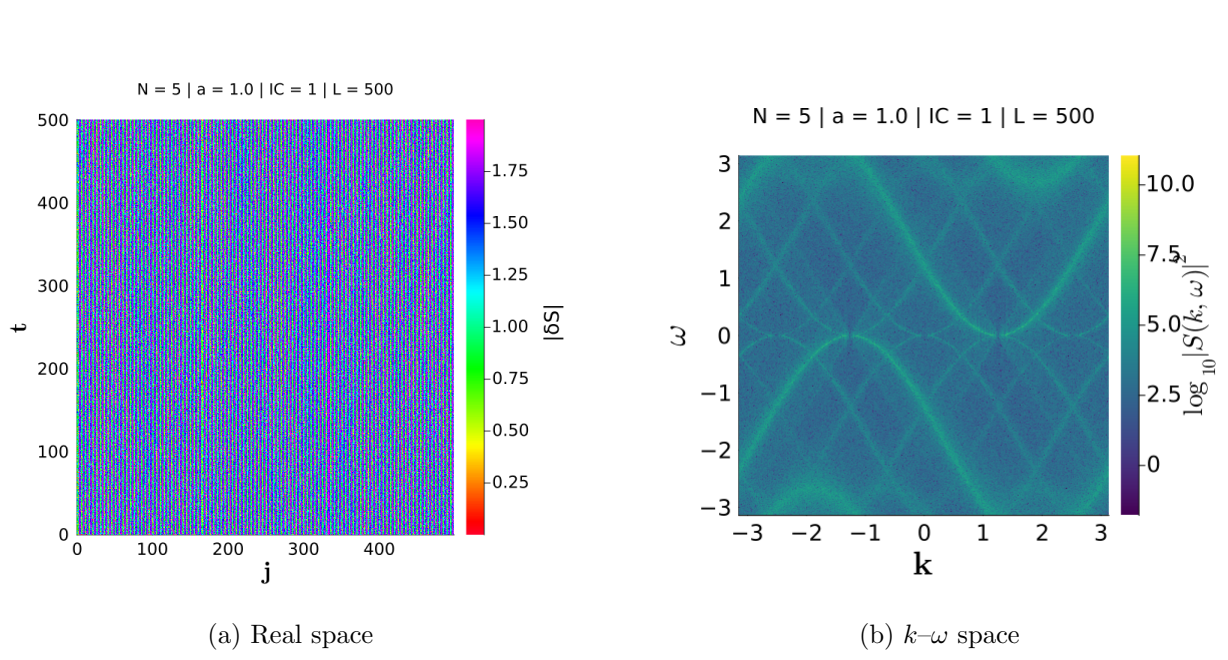


Figure 6: $N = 5$, $a = 1.0$, $L = 500$.

2 $N = 6$ ($L = 600$)

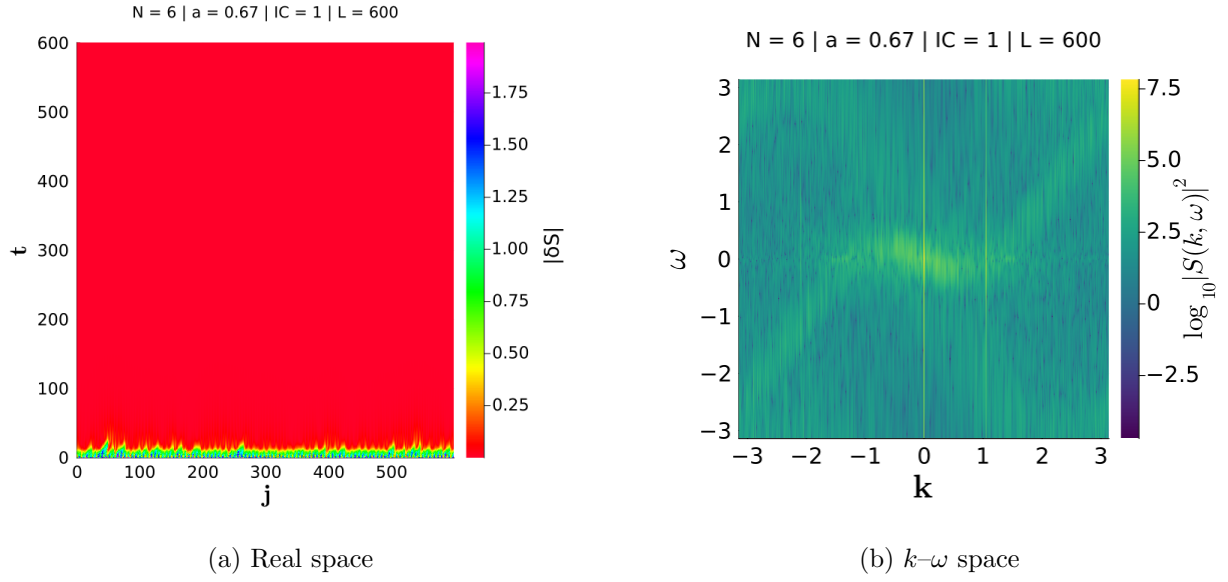


Figure 7: $N = 6$, $a = 0.67$, $L = 600$.

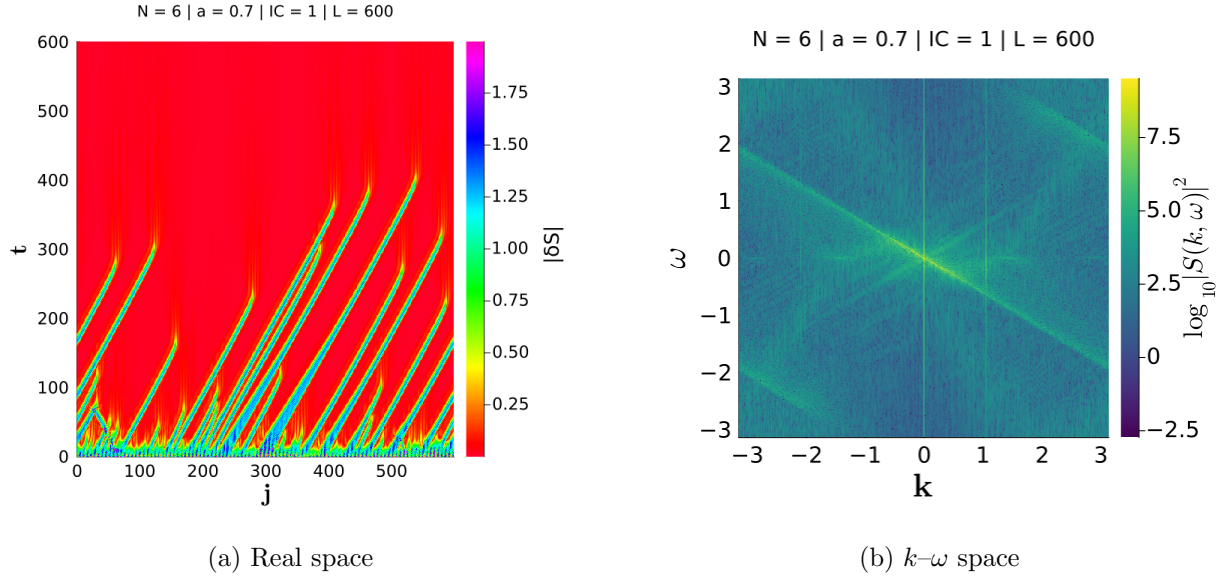
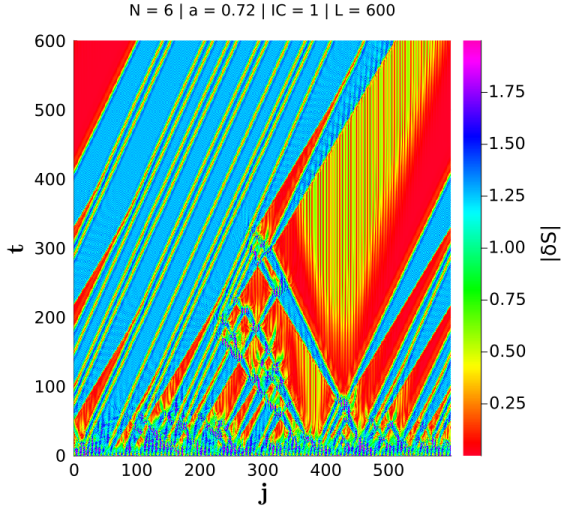
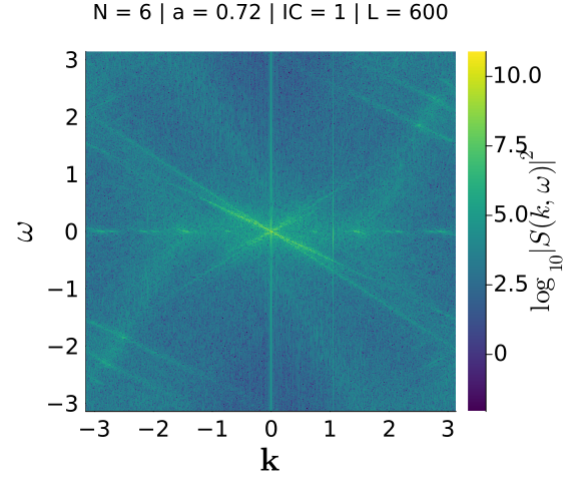


Figure 8: $N = 6$, $a = 0.7$, $L = 600$.

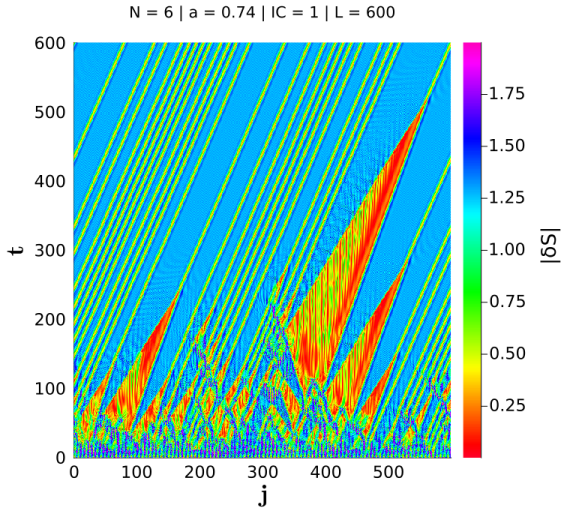


(a) Real space

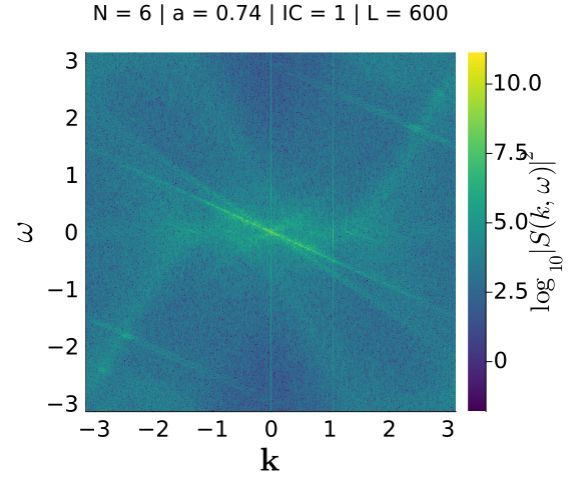


(b) k - ω space

Figure 9: $N = 6$, $a = 0.72$, $L = 600$.

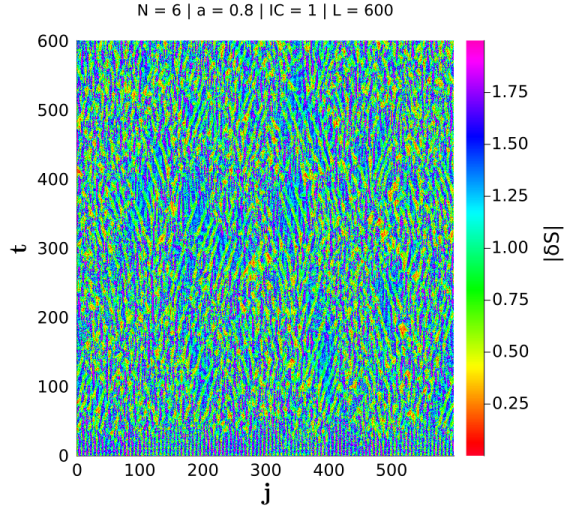


(a) Real space

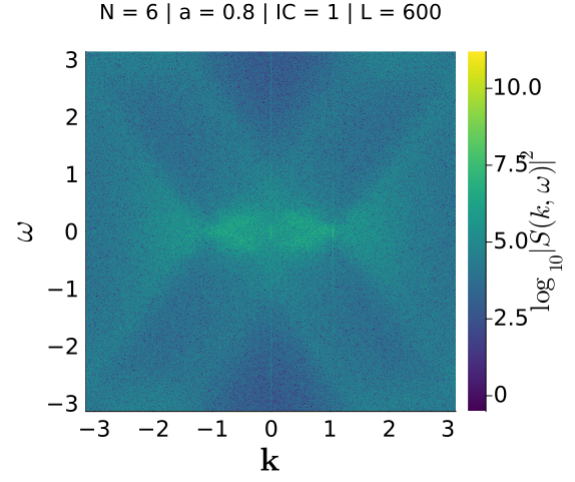


(b) k - ω space

Figure 10: $N = 6$, $a = 0.74$, $L = 600$.

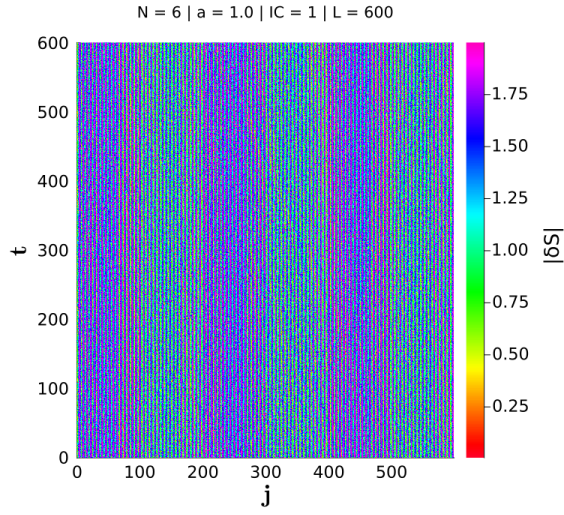


(a) Real space

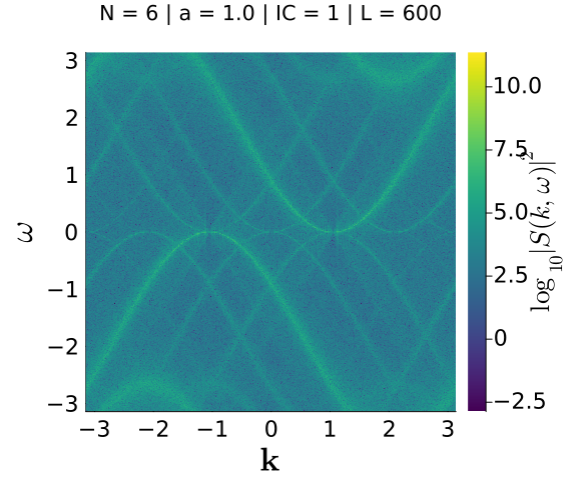


(b) k - ω space

Figure 11: $N = 6$, $a = 0.8$, $L = 600$.



(a) Real space



(b) k - ω space

Figure 12: $N = 6$, $a = 1.0$, $L = 600$.